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Understanding User Acceptance of AI-Driven Chatbots in China's E-Commerce: The Roles of Perceived Authenticity, Usefulness, and Risk

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Abstract: This study examines users' perceptions of Chatbots in China, with a particular focus on the factors influencing their acceptance and usage. Grounded in the Technology Acceptance Model (TAM), we analyze data from 542 online responses to explore the roles of Perceived Authenticity, usefulness, and risk in shaping user behavior toward AI-driven Chatbots. Using linear regression and mediation analyses, our findings indicate that both Perceived Authenticity and Perceived Usefulness positively impact users' behavioral intentions, while Perceived Risk has a negative influence. Notably, Perceived Usefulness serves as a mediator between behavioral intentions and both Perceived Authenticity and Perceived Risk. These results contribute to the growing body of research on AI and e-commerce by providing empirical evidence of the key factors affecting Chatbot adoption. The study offers valuable implications for developers and marketers, suggesting that enhancing Perceived Authenticity and usefulness while addressing Perceived Risks can improve user acceptance. These insights are particularly pertinent for AI practitioners aiming to refine Chatbot technology and expand its application across various sectors.

Keywords: AI; artificial intelligence; online customer service; Chatbots

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1. Introduction

Recent rapid advancements in artificial intelligence (AI) have sparked both excitement and concern about its increasing integration into everyday life [1], highlighting the growing influence of human–AI interactions. As AI continues to evolve, its role in shaping various industries, especially e-commerce, becomes more pronounced [2]. User acceptance of AI, including technologies like Chatbots, plays a crucial role in determining the success of AI adoption. While Chatbots are becoming a prominent application in the e-commerce industry, their widespread use is not without significant challenges.

AI, particularly through innovations in speech recognition, forms the backbone of Chatbot technology, enabling businesses to engage customers through natural, non-digital communication methods such as text and voice [3]. However, despite the increasing reliance on Chatbots in the e-commerce sector, substantial barriers to their acceptance remain, largely stemming from concerns over their functionality and effectiveness. While

Chatbots are becoming more prevalent in social media and messenger applications [4], they are still far from perfect in terms of user experience.

Among recent advances in the use of Chatbots in e-commerce is in the application of the technology to serve as an expert shopping aide to consumers [5]. As the chatbot functionality improves due to the accelerated learning curve supported by advances in AI and processing power, the potential of Chatbots to drive e-commerce indirectly, as well as directly, becomes apparent, agreed on by [6], who found that Chatbots were able to effectively serve as a green evangelist and influencer. Related research, including that of Sundjaja et al. [7], has found that disclosure, transparency, and meaningfulness [8] remain areas of concern for consumers when using Chatbots. While advances in security have been made, trust remains a significant issue, as consumers continue to express concerns regarding the ethical, secure, and privacy-compliant use of Chatbots in online transactions [9].

The application of the Technology Acceptance Model (TAM) is well established in analyzing the adoption of AI in e-commerce, and it is often used to assess user satisfaction with new technologies [10,11]. However, there are notable challenges to AI acceptance, particularly in the context of Chatbots, that need to be addressed for them to be more widely adopted. As AI in Chatbots advances, they must constantly adapt to evolving customer expectations and trends in human–computer interactions (HCIs), a process that remains a significant hurdle [12]. Studies show that communication with Chatbots often lacks the depth and emotional connection users experience in human–human interactions, which can lead to lower user engagement [13]. Additionally, the time-consuming nature of interactions with Chatbots, compounded by issues of response efficiency, poses significant barriers to user satisfaction and acceptance [14].

One of the primary challenges in Chatbot adoption is the issue of "productivity"—users expect fast, efficient responses, and when Chatbots fall short of these expectations, their use is resisted [12]. Moreover, concerns surrounding AI's intelligence, which is largely based on machine learning, are growing. As these technologies become more integrated into everyday life, so does awareness of their potential risks, such as social bias in AI interactions and the "creepiness" factor associated with AI's sometimes unpredictable outputs [15]. Research also highlights the role of anthropomorphic design elements in shaping user perceptions and increasing acceptance of Chatbots. For instance, Alboqami [11] emphasized that human-like characteristics in AI Chatbots, such as personalized interaction styles and emotional expressiveness, can enhance user trust and engagement, making them more likely to adopt the technology [16,17].

Despite these challenges, AI's potential to enhance user experience through improved HCI remains undeniable. However, in the e-commerce industry, particularly in China, the application of Chatbots faces unique challenges. China's online market, the largest globally, is characterized by a highly engaged user base that differs significantly from other regions, necessitating tailored strategies for AI acceptance [18,19]. The rapid growth of this market highlights both opportunities and challenges in integrating Chatbots into the shopping experience.

This study seeks to fill a gap in the current literature by addressing the challenges of AI acceptance in China's e-commerce sector, with a particular emphasis on AI-driven Chatbots. While prior studies have primarily focused on the technical capabilities and general adoption patterns of Chatbots [20–22], our research takes a distinct approach by examining the role of Perceived Authenticity as a new construct and investigating how Perceived Usefulness mediates its impact on user acceptance. This perspective allows us to uncover the nuanced mechanisms underlying user behavior, which have been overlooked in previous research. Specifically, we examine how Perceived Authenticityand Perceived Risk influence behavioral intention and how these relationships are mediated by Perceived

Usefulness. By highlighting these mediating mechanisms, our study provides a novel perspective on the complex dynamics that drive user behavior. Furthermore, by comparing and contrasting our findings with the existing literature, we deliver actionable insights for chatbot designers. These insights can inform strategies to refine chatbot features and interfaces, enhancing user satisfaction, increasing market demand, and improving the overall effectiveness of Chatbots in the e-commerce context. This focus on the mediating roles of Perceived Authenticity and Perceived Usefulness sets our study apart and underscores its unique contribution to advancing the understanding of AI-driven Chatbots. By understanding how different factors interact to shape user behavior, this research aims to provide actionable insights for designers of Chatbot systems. These insights can guide adjustments to Chatbot features and design, making them more attractive and effective for users, thus enhancing the market demand and overall user satisfaction in the e-commerce context.

This paper is structured as follows: Section 2 explores the theoretical underpinnings and hypothesis development, including a review of chatbot trends; the Technology Acceptance Model (TAM); and its relevance to behavioral intention, Perceived Authenticity, and Perceived Risk. Section 3 presents the conceptual framework, highlighting key constructs and their proposed relationships. Section 4 details the methodology, describing the survey instrument, sample demographics, and data collection process. Section 5 analyzes the results, utilizing regression and structural equation modeling to test the hypotheses. Finally, Section 6 discusses the findings, theoretical contributions, practical implications, and limitations, offering directions for future research on chatbot adoption and user behavior in the Chinese e-commerce context.

2. Literature Review, Hypotheses, and Conceptual Model Development

2.1. Chatbot Trends

Chatbots, as highly visible and significant implementations of AI technology, play a central role in both applied research and practical applications, particularly in natural language processing. Their efficiency and convenience have spurred advancements across various domains, resulting in sustained and widespread interest within the academic community [23]. AI has entered a new evolutionary phase known as AI 2.0, characterized by rapid developments due to reduced computing costs, an abundance of data, and enhanced algorithms [24]. With recent breakthroughs in natural language processing technologies, Chatbots have seen explosive growth, significantly increasing their user base through successful integration into smartphone applications, especially within social media platforms like WeChat, Facebook Messenger, and WhatsApp [25]. This visibility has not only boosted public awareness but also attracted substantial investment from prominent figures such as Hong Kong billionaire Li Ka-shing, Alibaba founder Jack Ma, and renowned technology investor Dr. Li Kaifu.

Beyond traditional uses in social media and e-commerce [26], Chatbots are being employed for increasingly sophisticated tasks such as Customer Relationship Management (CRM), where they enable highly personalized customer experiences [27]. Lorenzo et al. noted an emerging focus among online retailers on employing interactive technological solutions to enhance consumer engagement and improve efficiency and decision-making processes [28].

Interestingly, while many business sectors experienced downturns due to the COVID-19 pandemic, online activities in several sectors saw an upturn, notably in services like food delivery applications. Oncioiu [29] highlighted that Chatbots are increasingly utilized as decision-making aids, enabling retail managers to devise efficient and optimal logistics strategies in response to the norm of multichannel consumer behavior. In this context, recent research has indicated that the warmth of a chatbot's conversational style can

significantly reduce customers' Perceived Risks, more effectively than conversational competence, thereby enhancing trust in Chatbots [30].

When considering the acceptance of Chatbots and their overall functionality, it is crucial to acknowledge that, in human-to-machine (HTM) text communication, interactions often involve a mix of live human and machine responses, allowing users the option to choose direct human interaction if the robot's response proves unsatisfactory. Precisely, genuine chatbot–human interaction occurs only in communications without human intervention [31].

This section aims to provide a comprehensive overview of the current landscape of chatbot technology, highlighting their growing role in modern digital interactions and the ongoing research aimed at enhancing their development and integration into daily life.

2.2. Technology Acceptance Model

Methodologically, this study utilizes an adapted version of the Technology Acceptance Model (TAM), as proposed by Davis in 1989. The frequent adaptation of TAM, which includes the introduction of new factors or the omission of certain original elements, reflects the dynamic nature of technology and the influence of various contextual factors [32]. The TAM posits that user acceptance and the subsequent utilization of a technology are influenced by several key factors: Perceived Usefulness, perceived ease of use, and user attitude. Perceived ease of use is defined as the extent in which a person believes that using the technology will require minimal effort, while attitude encompasses the user's positive perceptions of the technology's benefits and ease of use [33].

The TAM is a well-established and widely recognized framework for assessing user acceptance and satisfaction. Its core variables, perceived ease of use and Perceived Usefulness, are well-documented antecedents that affect behavior and behavioral intentions [34]. This study expands the TAM framework by integrating context-specific variables such as Perceived Risk and Perceived Authenticity, which have been previously examined in studies on e-commerce acceptance [11,35]. Notably, this paper specifically omits the variables of perceived ease of use and attitude to concentrate on exploring the barriers to chatbot acceptance and the impacts of risk and authenticity. Previous research has indicated that attitude may have a negligible effect on technology usage, influenced by the methodological approach and other factors, especially in contexts where technology use is less discretionary [36–38]. As technology evolves, new applications for the TAM emerge; in this instance, the model is applied to the adoption of AI-driven Chatbots in mobile e-commerce within China.

2.3. Behavior Intention

Extensive research has been conducted to explore whether changes in behavioral intention directly influence people's actual behaviors [39] and has consistently identified behavioral intention as the primary determinant of overt behavior [40]. In this context, behavioral intention is often conceptualized as a dependent variable or an outcome that is influenced by various independent variables, such as perceived ease of use and Perceived Usefulness [41] or risk factors, as delineated in the Use & Gratification Model [42].

Behavioral intention also encompasses the notion of continuous consumption, which is described as a cognitive decision-making process [43]. The continuous intention to purchase typically develops after an initial usage or purchasing experience. These intentions are usually solidified following the consumption experience, which can lead customers to reuse the service or product or even recommend the experience to others [44]. Therefore, this paper will examine behavioral intention as a quantifiable indicator of consumers'

acceptance of Chatbots, providing insights into how these intentions translate into actual user engagement and loyalty.

2.4. Perceived Authenticity

2.4.1. Perceived Authenticity and Perceived Usefulness

According to Role Theory, consumers expect authenticity during interactions, which extends to virtual services and the concept of e-authenticity [45]. Authenticity plays a crucial role in shaping users' perceptions of the information they acquire. When interactions are perceived as authentic, users are more likely to find the information credible and relevant, which directly enhances Perceived Usefulness (PU)—users' evaluations of how effectively an interaction system meets their needs [46]. This perception of usefulness is central to users' continued engagement with the system.

Building on the Technology Acceptance Model (TAM), authenticity is linked to intrinsic motivators that influence behavioral beliefs, such as Perceived Usefulness [33,47]. When users perceive an interaction as authentic, their trust in the information increases, and they evaluate the interaction as more useful, thereby enhancing their overall perception of the system's utility.

Research in Electronic Word of Mouth (eWOM) supports this by demonstrating that authentic content increases users' trust and Perceived Usefulness of information. For example, in the context of e-commerce, authentic reviews not only increase trust in the platform but also enhance the Perceived Usefulness of the information provided, influencing users' perception of the system's effectiveness [48,49].

Additionally, from the perspective of Self-Determination Theory (SDT), environments that promote authenticity foster intrinsic motivation, which directly contributes to more positive evaluations of system utility [50]. Based on these theoretical perspectives, the following hypothesis is proposed:

H1: *Perceived Authenticity positively influences Perceived Usefulness*.

2.4.2. Perceived Authenticity and Perceived Behavior Intention

Perceived Authenticity has become a crucial factor in influencing consumer behavior, especially in online and retail environments. Research in the context of the Use & Gratification Model shows that authentic interactions—whether through human or AI communication—can significantly enhance trust and the perceived value, which, in turn, drive behavioral intentions, including repurchase and continued use [51,52]. Similarly, studies focused on the repurchase intentions of Airbnb users demonstrate that the Perceived Authenticityof information, particularly from AI-driven systems, positively influences user perceptions and increases the likelihood of repeat behavior [53]. These findings highlight the significant role of Perceived Authenticity in establishing trust and fostering emotional connections, both of which are critical in shaping consumer decisions.

In online commerce, where customers often interact with virtual entities rather than human representatives, the authenticity of the service or response plays a pivotal role in shaping consumer intentions. The ability to discern between real and virtual responses impacts consumers' willingness to engage, as Perceived Authenticity helps in fostering trust and emotional connection [46]. This connection, in turn, influences consumption intentions by enhancing the perceived friendliness and reliability.

The Self-Determination Theory (SDT) suggests that authenticity plays a key role in fostering intrinsic motivation, which is associated with more positive and genuine behavioral intentions. Similarly, the Technology Acceptance Model (TAM) emphasizes that authenticity can build trust and enhance perceived ease of use, both of which are essential

drivers of behavioral intentions [47]. Based on these theoretical foundations, the following hypothesis is proposed:

H2: *Perceived Authenticity positively influences behavioral intention.*

2.5. Perceived Risk

2.5.1. Perceived Risk and Perceived Usefulness

Extensive research has established a clear relationship between Perceived Risk and Perceived Usefulness, with Perceived Risk often acting as a significant barrier to technology adoption [54]. According to the Technology Acceptance Model (TAM), Perceived Risk is expected to negatively impact Perceived Usefulness, thereby influencing individuals' behavioral intentions to adopt new technologies [10]. Risk factors such as concerns about immature technology, privacy, security, and financial losses have been identified as particularly detrimental to Perceived Usefulness and overall user acceptance [42,55].

Perceived Risk has been categorized into various dimensions, including performance risk, financial risk, social risk, time risk, and psychological risk [56,57]. Among these, performance risk, defined as the likelihood that a product or service will not meet expectations, has been shown to have a particularly strong negative effect on Perceived Usefulness, especially in contexts where functional expectations are critical, such as online banking services [58]. This suggests that the perceived inability of a technology to deliver the expected performance can reduce users' perception of its usefulness. Given the established negative relationship between Perceived Risk and Perceived Usefulness, another hypothesis is formed:

H3: *Perceived Risk negatively influences Perceived Usefulness.*

2.5.2. Perceived Risk and Behavior Intention[1]

Risk is a central concept in human choice theory and has long been a key factor in understanding consumer behavior [59,60]. Since the 1960s, Perceived Risk has been a critical element in consumer choice research, influencing a wide range of behaviors [61]. Specifically, in the context of mobile banking, Perceived Risk plays a crucial role in shaping consumer attitudes toward the adoption of mobile banking apps, often exerting a negative influence on adoption [62]. Similarly, in studies of online application usage, Perceived Risk indirectly affects users' intentions by influencing their Perceived Usefulness and attitudes toward the application [63].

In the field of e-commerce, Perceived Risks—such as concerns about privacy breaches, data security, and the accuracy of chatbot interactions—pose significant barriers to user adoption and sustained engagement. While AI enhances customer behavior prediction and personalization, it also raises ethical concerns [64]. These risks can erode users' trust in chatbot-driven platforms, diminishing their perceived value and ultimately deterring behavioral intentions like continued use or recommendations. For instance, Featherman and Pavlou [65] highlighted that Perceived Risk negatively impacts users' perceptions of an e-service's usefulness and overall adoption intentions. Similarly, Zarouali et al. [66] demonstrated that trust and Perceived Risk are crucial factors influencing consumer responses to chatbot interactions on e-commerce platforms.

Moreover, Perceived Risk significantly influences consumer evaluations and behaviors, as trust and Perceived Authenticity are critical factors for adoption [67]. Similarly, Featherman and Pavlou [65] demonstrated that Perceived Risk negatively affects users' perceptions of an e-service's usefulness and their overall adoption intentions. These findings underscore the broad applicability of Perceived Risk across different consumer domains,

revealing its potential to shape behaviors and intentions. Based on these insights, the following hypothesis is proposed:

H4: *Perceived Risk negatively influences behavioral intention.*

2.5.3. Perceived Authenticity and Perceived Risk

Perceived Authenticity is increasingly recognized as a key factor influencing risk perception [53]. Consumers often rely on Perceived Authenticity to evaluate the trustworthiness and quality of a product, especially in environments where information about the product is scarce or uncertain. A lack of trustworthy information can lead consumers to question the authenticity of the product, which, in turn, elevates their Perceived Risk associated with purchasing it [68]. When consumers perceive a product or service as inauthentic, the uncertainty surrounding its value increases, which enhances the perception of risk.

Further research has highlighted that Perceived Authenticity and Perceived Risk are inversely related. A study utilizing a combination of qualitative and quantitative methods found that the perception of originality and the projection of authenticity are critical in mitigating Perceived Risk. The study concluded that as the Perceived Authenticity of a product or service increases, Perceived Risk decreases, emphasizing the protective role of authenticity in consumer decision-making [69].

This suggests that enhancing Perceived Authenticity can reduce consumer risk perceptions by assuring them of the product's quality, trustworthiness, and alignment with their expectations. As authenticity becomes more apparent, the likelihood of Perceived Risk diminishes, which, in turn, influences consumers' behavioral intentions toward purchasing or engaging with the product. Based on these insights, the following hypothesis is proposed:

H5: *Perceived Authenticity negatively influences Perceived Risk.*

2.6. Perceived Usefulness and Behavior Intention

Perceived Usefulness (PU) is defined as "the degree to which a person believes that using a particular system would enhance his or her job performance" [33]. It is a central concept in technology acceptance and consumer behavior research, representing the perceived value a user attributes to a system based on its ability to improve productivity, efficiency, or satisfaction. The relationship between Perceived Usefulness and behavioral intention has been widely studied, often highlighting the impact of PU on individuals' decision-making to adopt or continue using technology.

The Technology Acceptance Model (TAM), a widely accepted framework for understanding technology adoption, identifies Perceived Usefulness as a key predictor of behavioral intention [33,47]. In particular, TAM suggests that, when users perceive a technology to be useful, they are more likely to adopt or continue using it. This relationship has been consistently observed across different user demographics, including older adults who are particularly influenced by the perceived utility of a system [70,71].

Furthermore, numerous studies have shown a strong positive correlation between Perceived Usefulness and behavioral intention. For example, research examining professionals' perceptions of workplace technologies has found that PU strongly correlates with their intentions to adopt or use these technologies [72]. Similarly, in the context of mobile applications, the Perceived Usefulness of the app is a significant determinant of continued usage [73].

However, the relationship between Perceived Usefulness and behavioral intention is not universally consistent across all contexts. Some studies have reported varying or even negligible effects. For instance, research on social media websites found no significant cor-

relation between Perceived Usefulness and behavioral intention [74], while an e-commerce study concluded that Perceived Usefulness was not a significant predictor of Behavior Intention in their context [10]. These inconsistent findings suggest that the influence of Perceived Usefulness on behavioral intention may depend on specific factors, such as the technology's nature, user experience, and contextual relevance.

In light of these varying results, it is important to further explore the dynamics between Perceived Usefulness and behavioral intention in specific contexts. Given the growing use of Chatbots in the Chinese market, understanding this relationship in the context of Chinese consumers' adoption of chatbot technology can offer valuable insights. Based on the above discussions, the following hypothesis is proposed:

H6: *Perceived Usefulness positively influences Behavior Intention.*

2.7. Mediation Role of Perceived Usefulness

The role of Perceived Authenticity in shaping user behavior, particularly in online environments, has been well documented. Authenticity, when perceived by users, enhances trust, emotional connection, and the overall value of the interaction, all of which are critical drivers of behavioral intention [46,47]. However, the direct impact of Perceived Authenticity on behavioral intention may not fully account for the complex nature of this relationship.

Perceived Usefulness (PU), as defined by the Technology Acceptance Model (TAM), has been identified as a significant predictor of behavioral intention, representing the degree to which users believe that a technology or system will enhance their performance or fulfill their needs [33]. Given that authenticity enhances users' trust and emotional connection, it is reasonable to expect that these positive perceptions would contribute to the Perceived Usefulness of the system or service, thus influencing users' intention to use or adopt the technology.

Therefore, Perceived Usefulness may act as a mediator, translating the effect of Perceived Authenticity into tangible behavioral intentions. When users perceive a service or system as authentic, they are likely to find it more useful, which, in turn, strengthens their intention to engage with or adopt the system. This mediation relationship aligns with findings in the literature, where perceptions of trust, credibility, and emotional connection (all influenced by authenticity) directly contribute to how useful a system is perceived to be and how this Perceived Usefulness drives user behavior [49,75]. Based on these insights, the following hypothesis is proposed:

H7: *Perceived Usefulness mediates the relationship between Perceived Authenticity and behavioral intention.*

Perceived Risk has long been recognized as a key factor influencing consumer behavior, particularly in the context of technology adoption [59,60]. Consumers' perceptions of risks—such as concerns about privacy, security, performance failure, or financial loss—can significantly deter their intentions to engage with a new technology or service [62]. However, the impact of Perceived Risk on behavioral intention is not always direct; rather, it can be influenced by intermediate factors, such as Perceived Usefulness.

Perceived Usefulness plays a crucial role in mitigating the negative effects of Perceived Risk. When users perceive a technology or service as useful, they are more likely to overlook or tolerate certain risks, as the anticipated benefits outweigh potential drawbacks [63]. In other words, even when users perceive risks associated with a product or service, the Perceived Usefulness of the technology may reduce their reluctance and foster positive behavioral intentions. This is particularly relevant in contexts such as mobile banking or

mHealth applications, where concerns over privacy and security are prevalent but the Perceived Usefulness of the service may still drive users to adopt it [54,62].

Thus, Perceived Usefulness may mediate the relationship between Perceived Risk and behavioral intention by serving as a counterbalance to the negative effects of Perceived Risk. When users perceive that a system or service offers significant value, they may be more likely to proceed with the adoption, despite their concerns. This mediation effect aligns with previous research, which has shown that Perceived Usefulness can reduce the impact of Perceived Risk on users' adoption decisions [46]. Based on this rationale, the following hypothesis is proposed:

H8: *Perceived Usefulness mediates the relationship between Perceived Risk and behavioral intention.*

3. Conceptual Framework

Based on the comprehensive literature review, the pivotal factors identified that influence customer Behavior Intention are Perceived Risk, Perceived Usefulness, and Perceived Authenticity [76]. In this model, Perceived Usefulness is proposed as a mediator that channels the effects of Perceived Authenticity and Perceived Risk onto Behavior Intention. It is essential to assess the levels of these variables—Perceived Authenticity, Perceived Risk, and Perceived Usefulness—to clearly understand their interrelationships and their combined impact on user behavior.

Recent studies further support the importance of these factors in influencing user acceptance of Chatbots. For example, Xiao et al. [30] highlighted how chatbot conversation styles and human-like attributes impact user acceptance, perceived tie strength, and Perceived Risk. Their findings emphasize the significance of humanizing chatbot interactions to enhance user perception and reduce Perceived Risk, which aligns with the proposed model. Similarly, Bhaskar et al. [77] explored the interplay between perceived trust and Perceived Risk in the adoption of ChatGPT by educators, providing relevant insights into how these perceptions shape adoption intention. While this study is set in an educational context, its implications extend to understanding user behavior in other domains, including e-commerce.

Additionally, Silva et al. [78] examined strategies for mitigating Perceived Risk and improving user experience in e-retailing chatbot applications. Their research highlights the critical role of addressing Perceived Risk to foster a seamless user experience, thereby enhancing the effectiveness and acceptance of Chatbots in practical settings. These insights underscore the need to evaluate the interconnectedness of Perceived Authenticity, Perceived Risk, and Perceived Usefulness as proposed in our model.

The central research question of this study is: Will the use of Chatbots in China be negatively affected by the factors of Perceived Authenticity, Perceived Risk, and Perceived Usefulness? Figure 1 illustrates our conceptual model.

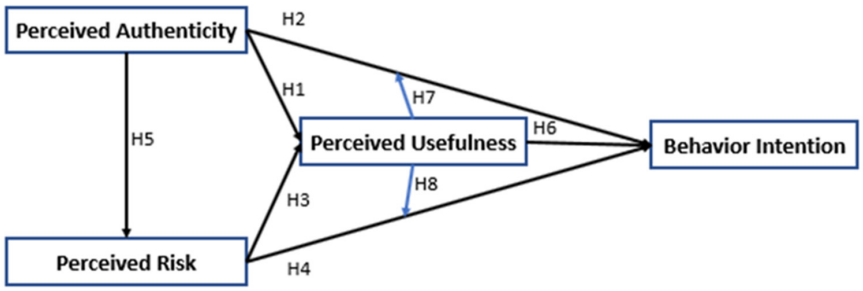


Figure 1. Conceptual model.

To address this question, the study will measure the aforementioned variables as outlined in the literature review. This approach will allow for an empirical examination of how these factors collectively influence the acceptance and effectiveness of Chatbots in the Chinese market, providing insights that could guide future developments in Chatbot technology.

4. Methodology

4.1. Sample and Data Collection

This study collected data using WeChat, a widely used social media platform that supports online surveys. The survey's QR code was distributed across various WeChat groups, a common practice in China and other regions for gathering data and sharing customer feedback or product reviews. Due to the high penetration of smartphones in China and the ubiquity of WeChat, which is integral to online shopping and e-commerce activities, nearly all users met the study's participation criteria.

A snowball sampling method was employed, which is typical for online surveys in China [79]. This approach yielded a satisfactory number of participants. A total of 542 usable responses were completed by individuals who self-identified as familiar with online shopping and Chatbots. Data collection occurred over a 14-day period. Participants were informed about the study's purpose and assured of their anonymity, with participation being voluntary. Since the data collected was anonymous and the study lacked external funding, the institutional policy did not require an Institutional Review Board (IRB) review.

The demographic characteristics of the sample are provided in Table 1. Among the respondents, 315 identified as female, 201 as male, and 26 as other or undeclared genders. The majority of participants (51.1%) were between the ages of 18 and 25, and 78.4% reported having a neutral or greater familiarity with Chatbots, making them eligible for inclusion in the target demographic.

Table 1. Demographics of the sample.

Variables	Subcategory	Frequency	Proportion
Gender	Male	201	0.371
	Female	315	0.581
	Others	26	0.048
Age	<18	40	0.074
	18–25	277	0.511
	26–35	52	0.096
	36–45	70	0.129
	>46	103	0.190
	<18	40	0.074
Familiarity	Very familiar	82	0.151
	Familiar	183	0.338
	Neutral	160	0.295
	Unfamiliar	93	0.172
	Not familiar at all	24	0.044

N = 542.

The descriptive statistics (Table 2) provide an overview of the constructs measured in this study: Perceived Authenticity (PA), Perceived Risk (PR), Perceived Usefulness (PU), and Behavioral Intention (BI). Participants rated Chatbots as moderately authentic (mean PA = 3.92, SD = 1.68) and perceived moderate risk (mean PR = 3.75, SD = 0.96). Perceived Usefulness (mean PU = 4.35, SD = 0.98) and Behavioral Intention (mean BI = 4.00, SD = 1.00) showed generally positive responses. Each construct was calculated as the mean of its survey items (e.g., PA1, PA2, and PA3 for PA), ensuring consistency with the survey

structure. The variability in the responses highlights the need to address diverse user preferences and concerns to improve chatbot design and user satisfaction.

Table 2. Descriptive statistics for the constructs.

	Mean	Standard Deviation	Min	Max
PA	3.92	1.68	1	7
PR	3.75	0.96	1	6.25
PU	4.35	0.98	1	6.67
BI	4.00	1.01	1	6.67

In the following empirical analysis, SEM was utilized to estimate the constructs PA, PR, PU, and BI. These SEM-based estimates were then used in linear regression to test the proposed hypotheses, extending the insights beyond latent variable relationships to include their direct effects.

4.2. Survey and Instrumentation

The use of an online survey tool for data collection offers several advantages, including low cost, fast data acquisition, participant anonymity, and reduced risk of bias, compared to in-person interviews [80].

The first section of the survey gathered demographic information and assessed participants' familiarity with Chatbots. The remaining sections of the survey were designed to measure four constructs: Perceived Authenticity (PA), Perceived Risk (PR), Perceived Usefulness (PU), and Behavior Intention (BI). Specifically, the construct of Perceived Authenticity was measured using three related questions, while Perceived Risk was assessed with four questions. Perceived Usefulness and Behavior Intention each included three questions. These scales measured both independent and dependent variables. The research model consists of four variables, with Behavior Intention as the terminal dependent variable.

The items in the survey were adapted from established scales used in the prior research of Rese et al. [42] to measure PA, PR, PU, and BI, which were, in turn, adapted from the scales used in the previous work of Rauschnabel et al. [81] for PR and Childers et al. [82], Ko et al. [83], and Venkatesh and Davis [47] for PU. Scales for the measurement of BI were adapted from the scales used by Moon and Kim [84] and Venkatesh et al. [85].

Since the original questions were in English, they required translation into Chinese. All survey questions were edited for clarity. To ensure the accuracy and validity of the translation, two native Chinese speakers with PhDs from Western universities reviewed the initial translation. The survey was then back-translated into English to verify the translation's fidelity. The translation and back-translation process is a widely accepted method in cross-language research to ensure accuracy.

All survey questions were answered using a 7-point Likert scale, with participants rating their agreement from "1" (strongly disagree) to "7" (strongly agree). The 7-point Likert scale is recognized as an effective and reliable tool for measuring attitudes and perceptions [71].

5. Analysis and Results

This study used various analytical tools and procedures to test the hypotheses and produce empirical evidence supporting the findings. To test the proposed hypotheses, we adopted a two-step approach. In Section 5. Analysis and Results, we used SPSS to conduct a series of linear regression analyses, examining the relationships between the variables and assessing the statistical significance of the proposed paths. Subsequently, in Section 6. Robustness Check, we utilized Python to test the hypotheses using structural

equation modeling (SEM). This approach allowed us to validate the findings from the linear regression analyses and ensure the robustness of the results through a more comprehensive modeling framework.

5.1. Reliability Analysis

As indicated in Table 3, the reliability analysis confirms that the data are sufficiently reliable for further analysis.

Table 3. Reliability analysis *.

Factor	Factor Loadings	Reliability	Variance Explained
Perceived Authenticity		0.615	22.07%
PA2	0.839		
PA3	0.783		
PA1	0.760		
Perceived Risks		0.810	17.70%
PR4	0.826		
PR1	0.825		
PR3	0.792		
PR2	0.783		
Perceived Usefulness		0.864	17.62%
PU3	0.838		
PU1	0.829		
PU2	0.786		
Behavior Intention		0.842	16.74%
BI2	0.824		
BI1	0.816		
BI3	0.812		
Overall		0.615	74.15%

* The statements of the factors are shown as follows: PA1: Chatting with a Chatbot is almost the same as chatting with a human being. PA2: The Chatbots sometimes give responses that are improper or embarrassing. PA3: I find it easy to communicate with Chatbots in a normal way. PR1: I am worried about the privacy of my personal data. PR2: Chatbots collect too much private information. PR3: "Chatbots" text recognition is not accurate. PR4: Chatbots do not understand inquiries. PU1: Chatbots are useful to me. PU2: Chatbots serve a useful purpose. PU3: The Chatbot is reported to provide customer support or training, personal assistance, or help with a particular task at hand. BI1: I will recommend others to use Chatbots. BI2: I intend to use Chatbots in the future. BI3: I will use more Chatbots on devices or applications in the future.

Perceived Authenticity (PA): All questions measuring the Perceived Authenticity factor loaded separately, with no questions removed. The total variance explained by this variable is 22.07%, and the reliability coefficient is 0.615, which is deemed adequate for this factor.

Perceived Risk (PR): Similarly, all questions measuring Perceived Risk loaded separately, with no questions removed. This factor explains 17.70% of the variance, with a reliability coefficient of 0.810, indicating good reliability.

Perceived Usefulness (PU): All questions measuring Perceived Usefulness also loaded separately, and none were removed. This variable explains 17.62% of the variance, with a reliability coefficient of 0.864, which is considered excellent.

Behavior Intention (BI): For Behavior Intention, all questions loaded separately, with no questions removed. The total variance explained by this factor is 16.74%, and the reliability coefficient is 0.842, also indicating high reliability.

The overall reliability of the data is satisfactory, with a coefficient of 0.615, and the total variance explained by the factor solution is 74.15%, well above the acceptable threshold for validity. The survey items for each construct have proven to be both valid and reliable, as

the responses for each group of questions were consistent. Therefore, the data are deemed suitable for further analysis.

5.2. Linear Regression Analysis

5.2.1. Linear Regression of Perceived Authenticity on Perceived Usefulness

As shown in Table 4, the regression analysis reveals a statistically significant relationship between Perceived Authenticity (PA) and Perceived Usefulness (PU), with a *p*-value less than 0.05. The model summary table indicates that the R-squared value is 0.187, meaning that 18.7% of the variation in Perceived Usefulness can be explained by Perceived Authenticity. The standardized coefficient beta value of 0.432 suggests that, for every one-unit increase in Perceived Authenticity, Perceived Usefulness increases by 0.432 units. This demonstrates a positive relationship between the two variables. Consequently, H1 is supported, as Perceived Authenticity has a significant positive impact on Perceived Usefulness.

Table 4. Regression analysis of PA on PU.

Independent Variables	Dependent Variable	R^2	Standardized Coefficients	*t*	Sig.	Impact
PA	PU	0.187	0.432	11.141	0	Significant

5.2.2. Linear Regression of Perceived Authenticity on Behavior Intention

As shown in Table 5, the regression analysis indicates a statistically significant relationship between Perceived Authenticity (PA) and Behavior Intention (BI), with a *p*-value less than 0.05. The model summary shows that the R-squared value is 0.190, meaning that 19.0% of the variation in Behavior Intention can be explained by Perceived Authenticity. The standardized coefficient beta value of 0.438 suggests that, for every one-unit increase in Perceived Authenticity, Behavior Intention increases by 0.438 units. This indicates a positive relationship between the two variables. Therefore, H2 is supported, as Perceived Authenticity has a significant positive impact on Behavior Intention.

Table 5. Regression analysis of PA on BI.

Independent Variables	Dependent Variable	R^2	Standardized Coefficients	*t*	Sig.	Impact
PA	BI	0.192	0.438	11.318	0	Significant

5.2.3. Linear Regression of Perceived Risk on Perceived Usefulness

As shown in Table 6, the regression analysis indicates a statistically significant negative relationship between Perceived Risk (PR) and Perceived Usefulness (PU), with a *p*-value less than 0.05. The model summary shows that the R-squared value is 0.131, meaning that 13.1% of the variation in Perceived Usefulness can be explained by Perceived Risk. The standardized coefficient beta value of −0.362 suggests that, for every one-unit increase in Perceived Risk, Perceived Usefulness decreases by 0.362 units. This indicates a negative relationship between the two variables.

Table 6. Regression analysis of PR on PU.

Independent Variables	Dependent Variable	R^2	Standardized Coefficients	*t*	Sig.	Impact
PR	PU	0.131	−0.362	−9.034	0	Significant

These findings align with prior research by Featherman and Pavlou [64], which highlighted the negative impact of Perceived Risk on Perceived Usefulness. Their study demonstrated that higher levels of Perceived Risk undermine users' perception of a system's usefulness by introducing concerns about security, privacy, or potential performance failures. By confirming this relationship, our study further supports the hypothesis (H3) that Perceived Risk has a significant negative impact on Perceived Usefulness.

5.2.4. Linear Regression of Perceived Risk on Behavior Intention

As shown in Table 7, the regression analysis reveals a statistically significant negative relationship between Perceived Risk (PR) and Behavior Intention (BI), with a *p*-value less than 0.05. The model summary indicates an R-squared value of 0.149, meaning that 14.9% of the variation in Behavior Intention is explained by Perceived Risk. The standardized coefficient beta value of −0.386 suggests that, for every one-unit increase in Perceived Risk, Behavior Intention decreases by 0.386 units. This demonstrates a negative relationship between Perceived Risk and Behavior Intention. Therefore, H4 is supported, as Perceived Risk has a significant negative effect on Behavior Intention.

Table 7. Regression analysis of PR on BI.

Independent Variables	Dependent Variable	R^2	Standardized Coefficients	*t*	Sig.	Impact
PR	BI	0.149	−0.386	−9.711	0	Significant

5.2.5. Linear Regression of Perceived Authenticity on Perceived Risk

As shown in Table 8, the regression analysis demonstrates a statistically significant negative relationship between Perceived Authenticity (PA) and Perceived Risk (PR), with a *p*-value less than 0.05. The model summary indicates an R-squared value of 0.216, meaning that 21.6% of the variation in Perceived Risk is explained by Perceived Authenticity. The standardized coefficient beta value of −0.464 suggests that, for every one-unit increase in Perceived Authenticity, Perceived Risk decreases by 0.464 units. This indicates a negative relationship between Perceived Authenticity and Perceived Risk. Therefore, H5 is supported, as Perceived Authenticity has a significant negative effect on Perceived Risk.

Table 8. Regression analysis of PA on PR.

Independent Variables	Dependent Variable	R^2	Standardized Coefficients	*t*	Sig.	Impact
PA	PR	0.216	−0.464	−12.18	0	Significant

5.2.6. Linear Regression of Perceived Usefulness on Perceived Behavior Intention

As shown in Table 9, the regression analysis reveals a significant positive relationship between Perceived Usefulness (PU) and Behavior Intention (BI), with a *p*-value less than 0.05. The model summary indicates an R-squared value of 0.279, which means that 27.9% of the variation in Behavior Intention is explained by Perceived Usefulness. The standardized coefficient beta value is 0.528, indicating that, for every one-unit increase in Perceived Usefulness, Behavior Intention increases by 0.528 units. This confirms a positive relationship between Perceived Usefulness and Behavior Intention.

Our findings align with established research, such as Venkatesh and Davis [47] and Venkatesh et al. [85]. These studies consistently demonstrated that PU is a significant predictor of BI, as users who perceive a technology to be useful are more likely to develop the intention to use or adopt it. Our results further corroborate this well-documented relationship, reinforcing the validity of PU as a key determinant of behavioral intention.

Therefore, H6 is supported, as Perceived Usefulness has a significant positive effect on Behavior Intention.

Table 9. Regression analysis of PU on BI.

Independent Variables	Dependent Variable	R^2	Standardized Coefficients	*t*	Sig.	Impact
PU	BI	0.279	0.528	14.457	0	Significant

5.3. Mediation Analysis

To test H7, which posits that Perceived Usefulness mediates the relationship between Perceived Authenticity and Behavior Intention, a mediated regression analysis was conducted. Similarly, to test H8, which hypothesizes that Perceived Usefulness mediates the relationship between Perceived Risk and Behavior Intention, a mediated regression analysis was performed. The results of these analyses are presented in Tables 10 and 11, respectively.

Table 10. Mediation analysis of PU between PA and BI.

	Effect	Standard Error	LLCI	ULCI	T Statistic	*p* Value
Total Effect	0.4357	0.0385	0.3601	0.5113	11.3175	0
Direct Effect	0.3438	0.0438	0.2031	0.3751	6.6046	0
Indirect Effect	0.1465	0.0279	0.0923	0.2042		

Table 11. Mediation analysis of PU between PR and BI.

	Effect	Standard Error	LLCI	ULCI	T Statistic	*p* Value
Total Effect	−0.3489	0.0359	−0.4194	−0.2783	−9.7108	0
Direct Effect	−0.2437	0.0415	−0.3251	−0.1622	−5.8755	0
Indirect Effect	−0.1052	0.0324	−0.1694	−0.0395		

5.3.1. Perceived Usefulness Between Perceived Authenticity and Behavior Intention

A mediated regression analysis was conducted to examine the mediating role of Perceived Usefulness (PU) in the relationship between Perceived Authenticity (PA) and Behavior Intention (BI) using PROCESS macro v4.0 by Andrew Hayes [86]. As shown in Table 10, the analysis provides significant support for the mediation effect. Specifically, Perceived Usefulness accounts for a 14.65% indirect effect from Perceived Authenticity (PA) to Behavior Intention (BI). This mediation effect is considered significant, as the confidence interval does not include zero. The lower limit of the confidence interval (LLCI) is 0.0923, and the upper limit (ULCI) is 0.2042, both of which exclude zero, confirming the presence of a significant indirect effect. Therefore, Perceived Usefulness mediates the relationship between Perceived Authenticity and Behavior Intention, and hypothesis H7 is supported.

5.3.2. Perceived Usefulness Between Perceived Risk and Behavior Intention

A mediated regression analysis was conducted to examine the mediating role of Perceived Usefulness (PU) in the relationship between Perceived Risk (PR) and Behavior Intention (BI) using PROCESS macro v4.0. As shown in Table 11, the results provide significant support for the mediation effect. Specifically, Perceived Usefulness accounts for a 10.52% indirect effect from Perceived Risk (PR) to Behavior Intention (BI). This mediation effect is considered significant, as the confidence interval does not include zero. The lower limit of the confidence interval (LLCI) is −0.1694, and the upper limit (ULCI) is −0.0395, both of which exclude zero, confirming the presence of a significant indirect effect.

Therefore, Perceived Usefulness mediates the relationship between Perceived Risk and Behavior Intention, and hypothesis H8 is supported.

6. Robustness Check

To test the proposed hypotheses (H1–H8), we conducted a structural equation modeling (SEM) analysis using the "semopy" library in Python. The results are presented in Table 12. The model comprised several structural relationships. First, Perceived Usefulness (PU) was modeled as a function of Perceived Authenticity (PA) and Perceived Risk (PR). Second, Behavioral Intention (BI) was defined as a function of PU, PA, and PR. Third, PR was modeled as a function of PA. Additionally, indirect effects were calculated to test the mediating role of PU in two relationships: (1) between PA and BI and (2) between PR and BI.

Table 12. The results of the SEM.

Hypothesis	Direct Effect	Indirect Effect	Total Effect
H1	0.1073 ($p < 0.05$)		
H2	0.0715 ($p < 0.05$)		
H3	−0.1994 ($p < 0.05$)		
H4	−0.1605 ($p < 0.05$)		
H5	−0.1447 ($p < 0.05$)		
H6	0.4101 ($p < 0.05$)		
H7	0.0715	0.044	0.1155
H8	−0.1605	−0.0818	−0.2423

The results of the structural equation modeling (SEM) analysis provided empirical support for all hypothesized relationships (H1–H8). The findings for each hypothesis are detailed below. First, Perceived Authenticity (PA) has a positive and statistically significant effect on Perceived Usefulness (PU) (Estimate = 0.1073, $p < 0.05$). This supports the hypothesis that individuals perceiving higher authenticity in Chatbots are more likely to find them useful. Therefore, H1 is supported.

Perceived Authenticity (PA) also positively influences Behavioral Intention (BI). The path coefficient for PA $\rightarrow$ BI is positive and statistically significant (Estimate = 0.0715, $p < 0.05$), indicating that perceptions of authenticity directly enhance the likelihood of recommending or using Chatbots in the future. Thus, H2 is supported.

Perceived Risk (PR) negatively influences Perceived Usefulness (PU). The SEM analysis reveals a significant negative relationship between PR, measured by concerns about privacy (PR1 and PR2) and system inaccuracies (PR3 and PR4), and PU (Estimate = −0.1994, $p < 0.05$). This confirms that a higher Perceived Risk reduces the Perceived Usefulness of Chatbots. Hence, H3 is supported.

A significant negative relationship was observed between Perceived Risk (PR) and Behavioral Intention (BI). The path coefficient for PR $\rightarrow$ BI is negative and statistically significant (Estimate = −0.1605, $p < 0.05$), suggesting that concerns about chatbot privacy and performance undermine the intention to use or recommend Chatbots. Accordingly, H4 is supported.

The SEM analysis indicates that Perceived Authenticity (PA) negatively influences Perceived Risk (PR) (Estimate = −0.1447, $p < 0.05$). This finding supports the hypothesis that higher Perceived Authenticity reduces Perceived Risks associated with Chatbots. Therefore, H5 is supported.

Perceived Usefulness (PU) positively influences Behavioral Intention (BI). The path coefficient for PU $\rightarrow$ BI is positive and statistically significant (Estimate = 0.4101, $p < 0.05$), confirming that the usefulness of Chatbots strongly drives the intention to use or recommend them. Hence, H6 is supported.

The mediating role of Perceived Usefulness (PU) in the relationship between Perceived Authenticity (PA) and Behavioral Intention (BI) was also supported. The indirect effect of PA → PU → BI was calculated as $0.1073 \times 0.4101 = 0.0440$, indicating a statistically significant mediation effect. Thus, H7 is supported.

Similarly, Perceived Usefulness (PU) mediates the relationship between Perceived Risk (PR) and Behavioral Intention (BI). The indirect effect of PR → PU → BI was calculated as $-0.1994 \times 0.4101 = -0.0818$, confirming the mediating role of Perceived Usefulness. Consequently, H8 is supported.

The findings demonstrate that all direct and indirect hypotheses (H1–H8) were supported by the SEM analysis. Direct effects such as PA → PU, PU → BI, and PR → BI were statistically significant, while the calculated indirect effects confirmed the mediating role of PU in the relationships of PA → BI and PR → BI. These results provide robust empirical evidence for the hypothesized relationships. The SEM results generally show smaller effect sizes compared to separate regression results (from Table 4 to Table 11) due to its holistic approach, accounting for all relationships simultaneously.

The results of the hypotheses testing using linear regression and structural equation modeling are summarized in Table 13. The results from SEM align closely with separate regression outcomes, supporting the validity of the findings across both methods. The minor differences in effect sizes are expected and reflect the methodological distinctions between the two approaches.

Table 13. Hypotheses testing results.

	Hypothesis	Result
H1	Perceived Authenticity will positively influence the Perceived Usefulness.	Supported
H2	Perceived Authenticity positively influences Behavior Intention.	Supported
H3	Perceived Risk negatively influences Perceived Usefulness.	Supported
H4	Perceived Risk negatively influences Behavior Intention.	Supported
H5	Perceived Authenticity negatively influences Perceived Risk.	Supported
H6	Perceived Usefulness positively influences Behavior Intention.	Supported
H7	Perceived Usefulness mediates the relationship between Perceived Authenticity and Behavior Intention.	Supported
H8	Perceived Usefulness mediates the relationship between Perceived Risk and Behavior Intention.	Supported

7. Discussion

This study investigates users' perceptions of AI-driven Chatbots in China, focusing on the factors influencing their acceptance and usage within the context of e-commerce. Grounded in the Technology Acceptance Model (TAM), we analyzed data collected from 542 online survey responses to explore the roles of Perceived Authenticity, Perceived Usefulness, and Perceived Risk in shaping user behavior. By employing both linear regression and structural equation modeling (SEM), the study examined the direct and mediated effects of these factors on behavioral intention.

The findings reveal that Perceived Authenticity and Perceived Usefulness positively influence users' behavioral intentions (H2 and H6), while Perceived Risk exerts a negative impact (H4). Notably, Perceived Usefulness serves as a key mediator (H7 and H8), channeling the effects of both Perceived Authenticity and Perceived Risk onto behavioral intention. These results provide robust empirical evidence supporting the conceptual

model, confirming the significance of authenticity in fostering trust and reducing Perceived Risks and highlighting the central role of usefulness in driving user acceptance.

By bridging theoretical insights with practical applications, this study contributes to the growing body of research on AI and e-commerce. It emphasizes the need for developers and marketers to enhance Perceived Authenticity and usefulness while addressing concerns about risk to improve user acceptance. These findings are particularly relevant for practitioners aiming to refine chatbot technology, suggesting that user-centric designs and secure, practical functionalities are essential for broadening chatbot adoption across various sectors. Ultimately, the study provides actionable insights to guide future developments in AI-driven Chatbots and their application in diverse contexts.

7.1. Practical Implications

The increasing use of AI, particularly through Chatbots, is transforming the service and e-commerce sectors by bridging the gap left by the absence of personal, human-like interactions in digital environments. E-commerce offers significant advantages over traditional brick-and-mortar stores, yet it often lacks the human touch that builds consumer trust and satisfaction. Chatbots can address this challenge by simulating personalized and meaningful interactions between sellers and consumers, enhancing the overall e-commerce experience [22].

Our findings suggest that Perceived Authenticity is a crucial factor in shaping consumer perceptions of AI interactions. While AI continues to advance, the ability to engage users in authentic, emotionally resonant interactions remains a challenge. The results of this study indicate that higher levels of Perceived Authenticity can reduce the perceived emotional distance between human and machine interactions, diminishing the risks associated with AI technology. These results align with recent findings by Yazan and Situmeang [30], who emphasized the importance of human-like interaction styles in reducing Perceived Risk and enhancing Chatbot acceptance. By integrating such insights into chatbot design, developers can create more relatable and effective tools for consumers.

For Chatbot developers and marketers, these findings offer valuable insights for designing more effective AI systems. The most critical factor driving consumer behavioral intentions is Perceived Usefulness, which can significantly influence how consumers view Chatbots. As our research highlights, Perceived Usefulness can also mediate the negative effects of Perceived Authenticity and Perceived Risk. This finding resonates with Bhaskar et al. [77], who demonstrated that perceived trust and usefulness are key mediators in mitigating the impact of Perceived Risk on adoption intention. Marketers should prioritize understanding the drivers of Perceived Usefulness, ensuring that Chatbots are not only authentic but also practically beneficial to users. This involves focusing on enhancing functional capabilities over aesthetics, as consumers are more likely to engage with and adopt Chatbots that deliver clear, tangible value rather than those that merely appear appealing. Ultimately, this research underscores the importance of optimizing Chatbots for pragmatic, efficient use, which will lead to greater consumer satisfaction and sustained adoption.

7.2. Policy Implications

This study identifies Perceived Risk as a primary negative factor affecting consumer adoption of AI-driven Chatbots in e-commerce. Specifically, concerns regarding security and privacy significantly influence consumers' perceptions of the usefulness of these technologies, ultimately impacting their willingness to adopt them. Based on the findings of this study, the following policy implications are proposed.

First, there is a critical need for governments to strengthen privacy and data security regulations within the e-commerce sector. As consumers frequently share sensitive personal and financial data during online transactions, governments must establish and enforce clear, stringent guidelines to protect users' privacy. These measures should include explicit restrictions on the use and sharing of consumer data without consent. Such regulatory frameworks not only align with consumer expectations but also mitigate legal risks for companies. By prioritizing robust data protection policies, governments can help create a safer digital environment that encourages consumer adoption of chatbot technologies.

Second, the findings underscore the importance of investing in technological advancements to reduce Perceived Risks. E-commerce companies should focus on developing and implementing enhanced security mechanisms, such as advanced encryption methods and multi-layered information firewalls, to prevent data breaches and cyberattacks. The recent work by Silva et al. [78] underscored the role of robust security measures in reducing Perceived Risks and improving user experience, further highlighting the importance of secure infrastructure in chatbot implementation.

Lastly, the rapid expansion of e-commerce presents an opportunity to leverage chatbot technologies to drive socio-economic development. By fostering consumer trust in secure chatbot platforms, e-commerce can serve as a powerful tool for economic growth, particularly in underserved regions. Policies promoting the adoption of AI technologies in rural areas can help bridge the digital divide and enhance access to essential goods and services, thereby supporting national socio-economic goals. For example, during the COVID-19 pandemic, secure e-commerce platforms played a vital role in delivering healthcare and essential goods to remote populations. Governments should prioritize initiatives that promote the adoption of chatbot technologies, ensuring equitable access to digital services and fostering economic inclusion.

In summary, these policy recommendations—centered on mitigating Perceived Risk, enhancing security, and promoting equitable access—align directly with the empirical findings of this study. By addressing these factors, stakeholders can support the sustainable growth of AI-driven e-commerce platforms while contributing to broader socio-economic development goals.

7.3. Theoretical Contribution

The theoretical implications of the hypotheses and relationships discussed above are significant for the understanding of consumer behavior in the contexts of technology adoption and online interactions. Firstly, the concept of Perceived Authenticity plays a pivotal role in shaping users' perceptions and behaviors. The theories, such as Role Theory, Technology Acceptance Model (TAM), and Self-Determination Theory (SDT), highlight that, when consumers perceive an interaction or technology as authentic, it enhances their trust, emotional connection, and evaluation of the technology's usefulness. This perception of authenticity thus becomes a powerful antecedent to both Perceived Usefulness (PU) and behavioral intention (BI). By incorporating these theoretical perspectives, the findings suggest that authenticity influences users' intrinsic motivation and their evaluations of a system's utility, providing important theoretical insights into how trust and emotional engagement drive technology adoption.

Additionally, the relationship between Perceived Risk and Perceived Usefulness adds another layer of complexity. The Technology Acceptance Model (TAM) suggests that Perceived Risk, especially concerns over privacy, security, and performance failure, negatively affects the Perceived Usefulness of a system. However, as Perceived Usefulness mediates the relationship between Perceived Risk and behavioral intention, this highlights the counterbalancing role of usefulness in the face of risk. This complements the findings of

Bhaskar et al. [77], who demonstrated the centrality of perceived trust and usefulness in moderating the effects of Perceived Risk on adoption behaviors.

The incorporation of Perceived Usefulness as a mediator provides valuable theoretical insights into the risk mitigation process, showing that, even when users perceive risks, the anticipated benefits of a technology or service may override those concerns. This extends existing models by emphasizing the indirect paths through which Perceived Risk influences consumer behavior and suggesting that Perceived Usefulness not only enhances trust but also acts as a buffer against Perceived Risk, ultimately shaping behavioral intentions toward technology adoption and usage.

7.4. Limitations and Future Research

This study is subject to several limitations that should be addressed in future research. First, the generalizability of the findings is limited due to the sample's demographic composition, as the majority of respondents were Chinese individuals within a specific age range. To enhance the robustness and applicability of the results, future studies should consider diverse demographic groups, including various age cohorts, education levels, occupations, and cultural backgrounds. Expanding the sample will allow for a more comprehensive understanding of how different populations perceive and interact with Chatbot technologies.

Additionally, this study relies on an adaptation of the Technology Acceptance Model (TAM), a well-established theoretical framework. While TAM provides valuable insights, future research could explore the applicability of other related theories, such as the Unified Theory of Acceptance and Use of Technology (UTAUT) or the Task-Technology Fit Model, to uncover additional factors influencing Chatbot acceptance. Employing alternative theoretical frameworks might yield new perspectives and enhance the understanding of user behaviors toward Chatbots.

Another limitation arises from the specific sample of college students, which may not fully capture the diverse external factors that influence Chatbot adoption. Educational background, for example, could play a role in how individuals perceive and interact with AI-driven systems. Future studies could broaden the participant pool to include professionals from different industries and educational backgrounds, offering a more comprehensive view of Chatbot usage across various sectors.

The reliance on online surveys as the primary data collection method also introduces the possibility of response biases, such as social desirability bias or untruthful reporting. To mitigate this, future research could employ alternative data collection methods, such as semi-structured interviews, focus groups, or the analysis of online reviews. These approaches would provide richer, more nuanced insights and help validate the findings of this study.

Furthermore, expanding the scope of Chatbot applications beyond e-commerce could yield a broader understanding of the technology's potential. For example, examining Chatbots in healthcare, education, or customer service could highlight unique challenges and benefits across various industries. This would contribute to a more holistic view of Chatbot technology's impact on consumer behavior and society.

A mixed-methods approach that combines qualitative interviews with quantitative surveys could offer deeper insights into the factors driving Chatbot acceptance. Such an approach would allow researchers to explore not only the statistical relationships but also the underlying motivations and concerns of users. Additionally, investigating how user perceptions of Chatbots have evolved before and after the COVID-19 pandemic could reveal shifts in usage patterns and acceptance, providing valuable context for understanding the long-term impact of the pandemic on technology adoption.

Finally, another limitation of this study is the exclusion of the chatbot's hedonic aspect, such as enjoyment, entertainment value, and emotional engagement—factors that are increasingly recognized as critical in technology adoption. Due to time constraints, including the 10-day revision period for this manuscript, we were unable to integrate hedonic aspects into our survey and analysis. Future research could explore the interplay between utilitarian and hedonic factors in shaping user acceptance of and satisfaction with Chatbots. By investigating how hedonic attributes influence user perceptions alongside utilitarian constructs, subsequent studies could offer a more comprehensive understanding of user behavior and preferences. Such an extension would not only deepen the theoretical discourse but also provide actionable insights for developers to create Chatbots that are both functional and emotionally engaging.

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Notes

[1] Shapiro–Wilk tests were conducted for the residuals of all the regression models. No significant deviations from normality were observed in any case ($p > 0.05$).

References

1. Tian, W.; Ge, J.; Zhao, Y.; Zheng, X. AI Chatbots in Chinese higher education: Adoption, perception, and influence among graduate students—An integrated analysis utilizing UTAUT and ECM models. *Front. Psychol.* **2024**, *15*, 1268549. [CrossRef] [PubMed]
2. Rahevar, M.; Darji, S. The adoption of AI-driven Chatbots into a recommendation for e-commerce systems to targeted customer in the selection of product. *Int. J. Manag. Econ. Commer.* **2024**, *1*, 128–135. [CrossRef]
3. Shawar, B.A.; Atwell, E. Chatbots: Are they really useful? *J. Lang. Technol. Comput. Linguist.* **2007**, *22*, 29–49. [CrossRef]
4. Valério, F.A.; Guimarães, T.G.; Prates, R.O.; Candello, H. Here's what I can do: Chatbots' strategies to convey their features to users. In Proceedings of the XVI Brazilian Symposium on Human Factors in Computing Systems, Joinville, Brazil, 23–27 October 2017; pp. 1–10. [CrossRef]
5. Maruthi, P.B.; Prasad, D.; Niveditha, B. Chatbot in Ecommerce. In *Design and Development of Emerging Chatbot Technology*; IGI Global Scientific Publishing: Hershey, PA, USA, 2024; pp. 256–279. [CrossRef]
6. Sadiq, M.W.; Akhtar, M.W.; Huo, C.; Zulfiqar, S. ChatGPT-Powered Chatbot as a Green Evangelist: An Innovative Path toward Sustainable Consumerism in E-Commerce. *Serv. Ind. J.* **2024**, *44*, 173–217. [CrossRef]
7. Sundjaja, A.M.; Utomo, P.; Colline, F. The Determinant Factors of Continuance Use of Customer Service Chatbot in Indonesia E-Commerce: Extended Expectation Confirmation Theory. *J. Sci. Technol. Policy Manag.* **2024**, *16*, 182–203. [CrossRef]
8. Altay, B.C.; Çetintürk, N. Customer Dissatisfaction Towards Chatbot Services of E-Commerce Shopping Sites: A Qualitative Analysis. *J. Transp. Logist.* **2024**, *9*, 112–120. [CrossRef]
9. Thapliyal, K.; Thapliyal, M. Chatbot-XAI—The New Age Artificial Intelligence Communication Tool for E-Commerce. In *Role of Explainable Artificial Intelligence in E-Commerce*; Gaur, L., Abraham, A., Eds.; Springer Nature Switzerland: Cham, Switzerland, 2024; pp. 77–100. [CrossRef]
10. Alraja, M.; Mohammed, A. Customer acceptance of e-commerce: Integrating Perceived Risk with TAM. *Int. J. Appl. Bus. Econ. Res.* **2015**, *13*, 913–921.

11. Valencia, D.C.; Alejandro, V.-A.; Bran, L.; Benjumea, M.; Valencia, J. Analysis of e-commerce acceptance using the technology acceptance model. In *Scientific Papers of the University of Pardubice. Series D, Faculty of Economics and Administration*; Univerzita Pardubice: Pardubice, Czech Republic, 2019; Volume 45, Available online: https://hdl.handle.net/10195/72226 (accessed on 11 October 2024).
12. Brandtzaeg, P.B.; Følstad, A. Chatbots: Changing user needs and motivations. *Interactions* **2018**, *25*, 5. [CrossRef]
13. Hill, J.; Ford, W.R.; Farreras, I.G. Real conversations with artificial intelligence: A comparison between human–human online conversations and human–chatbot conversations. *Comput. Hum. Behav.* **2015**, *49*, 245–250. [CrossRef]
14. Pillai, R.; Sivathanu, B. Adoption of AI-based Chatbots for hospitality and tourism. *Int. J. Contemp. Hosp. Manag.* **2020**, *32*, 3199–3226. [CrossRef]
15. Rajaobelina, L.; Prom Tep, S.; Arcand, M.; Ricard, L. Creepiness: Its antecedents and impact on loyalty when interacting with a chatbot. *Psychol. Mark.* **2021**, *38*, 2339–2356. [CrossRef]
16. Gomes, S.; Lopes, J.M.; Trancoso, T. Customer experience in digital transformation: The influence of intelligent Chatbots toward a sustainable market. *Int. J. Innov. Sci.*. [CrossRef]
17. Alboqami, H. Factors affecting consumers' adoption of AI-based Chatbots: The role of anthropomorphism. *Am. J. Ind. Bus. Manag.* **2023**, *13*, 4. [CrossRef]
18. Farooq, Q.; Fu, P.; Hao, Y.; Jonathan, T.; Zhang, Y. A review of management and importance of e-commerce implementation in service delivery of private express enterprises of China. *Sage Open* **2019**, *9*, 2158244018824194. [CrossRef]
19. Su, W.; Wang, Y.; Qian, L.; Zeng, S.; Baležentis, T.; Streimikiene, D. Creating a sustainable policy framework for cross-border e-commerce in China. *Sustainability* **2019**, *11*, 943. [CrossRef]
20. Dale, R. The return of the Chatbots. *Nat. Lang. Eng.* **2016**, *22*, 811–817. [CrossRef]
21. Kaczorowska-Spychalska, D. How Chatbots influence marketing. *Management* **2019**, *23*, 251–270. [CrossRef]
22. Marjerison, R.K.; Zhang, Y.; Zheng, H. AI in E-Commerce: Application of the Use and Gratification Model to the Acceptance of Chatbots. *Sustainability* **2022**, *14*, 14270. [CrossRef]
23. Lin, C.C.; Huang, A.Y.; Yang, S.J. A review of ai-driven conversational Chatbots implementation methodologies and challenges. *Sustainability* **2023**, *15*, 4012. [CrossRef]
24. Zhang, D.; Han, X.; Deng, C. Review on the research and practice of deep learning and reinforcement learning in smart grids. *CSEE J. Power Energy Syst.* **2018**, *4*, 362–370. [CrossRef]
25. Io, H.N.; Lee, C.B. Chatbots and conversational agents: A bibliometric analysis. In Proceedings of the 2017 IEEE International Conference on Industrial Engineering and Engineering Management, Singapore, 10–13 December 2017; pp. 215–219. [CrossRef]
26. Galitsky, B.; Galitsky, B. Chatbots for CRM and dialogue management. In *Artificial Intelligence for Customer Relationship Management: Solving Customer Problems*; Springer: Cham, Switzerland, 2021; pp. 1–61. [CrossRef]
27. Guerola-Navarro, V.; Gil-Gomez, H.; Oltra-Badenes, R.; Soto-Acosta, P. Customer relationship management and its impact on entrepreneurial marketing: A literature review. *Int. Entrep. Manag. J.* **2022**, *20*, 507–547. [CrossRef]
28. Lorenzo-Romero, C.; Andrés-Martínez, M.-E.; Cordente-Rodríguez, M.; Gómez-Borja, M.Á. Active Participation of E-Consumer: A Qualitative Analysis From Fashion Retailer Perspective. *SAGE Open* **2021**, *11*, 2158244020979169. [CrossRef]
29. Lin, Y.; Marjerison, R.K.; Choi, J.; Chae, C. Supply Chain Sustainability during COVID-19: Last Mile Food Delivery in China. *Sustainability* **2022**, *14*, 1484. [CrossRef]
30. Xiao, R.; Yazan, M.; Situmeang, F.B.I. Rethinking Conversation Styles of Chatbots from the Customer Perspective: Relationships between Conversation Styles of Chatbots, Chatbot Acceptance, and Perceived Tie Strength and Perceived Risk. *Int. J. Hum.-Comput. Interact.* **2024**, 1–21. [CrossRef]
31. Melián-González, S.; Gutiérrez-Taño, D.; Bulchand-Gidumal, J. Predicting the intentions to use Chatbots for travel and tourism. *Curr. Issues Tour.* **2021**, *24*, 2. [CrossRef]
32. Moura, I.V.; de Almeida, L.B.; da Silva, W.V.; da Veiga, C.P.; Costa, F. Predictor Factors of Intention to Use Technological Resources: A Multigroup Study About the Approach of Technology Acceptance Model. *SAGE Open* **2020**, *10*, 2158244020967942. [CrossRef]
33. Davis, F.D. Perceived Usefulness, Perceived Ease of Use, and User Acceptance of Information Technology. *MIS Q.* **1989**, *13*, 319–340. [CrossRef]
34. Granic, I.; Lobel, A.; Engels, R.C.M.E. The benefits of playing video games. *Am. Psychol.* **2014**, *69*, 66–78. [CrossRef]
35. Handoyo, S. Purchasing in the digital age: A meta-analytical perspective on trust, risk, security, and e-WOM in e-commerce. *Heliyon* **2024**, *10*, e29714. [CrossRef]
36. Teo, T. Is there an attitude problem? Reconsidering the role of attitude in the TAM. *Br. J. Educ. Technol.* **2009**, *40*, 1139–1141. [CrossRef]
37. Thompson, R.L.; Higgins, C.A.; Howell, J.M. Personal computing: Toward a conceptual model of utilization. *MIS Q.* **1991**, *15*, 125–143. [CrossRef]
38. López-Bonilla, L.M.; López-Bonilla, J.M. Explaining the discrepancy in the mediating role of attitude in the TAM. *Br. J. Educ. Technol.* **2017**, *48*, 940–949. [CrossRef]

39. Lee, M. Factors influencing the adoption of internet banking: An integration of TAM and TPB with Perceived Risk and perceived benefit. *Electron. Commer. Res. Appl.* **2009**, *8*, 3. [CrossRef]
40. Wilson, D.T.; Mathews, H.L.; Harvey, J.W. An Empirical Test of the Fishbein Behavioral Intention Model. *J. Consum. Res.* **1975**, *1*, 39–48. [CrossRef]
41. Abdullah, F.; Ward, R.; Ahmed, E. Investigating the influence of the most commonly used external variables of TAM on students' Perceived Ease of Use (PEOU) and Perceived Usefulness (PU) of e-portfolios. *Comput. Hum. Behav.* **2016**, *63*, 75–90. [CrossRef]
42. Rese, A.; Ganster, L.; Baier, D. Chatbots in retailers' customer communication: How to measure their acceptance? *J. Retail. Consum. Serv.* **2020**, *56*, 102176. [CrossRef]
43. Colgate, M.; Lang, B. Switching barriers in consumer markets: An investigation of the financial services industry. *J. Consum. Mark.* **2001**, *18*, 332–347. [CrossRef]
44. Lin, H.F. Understanding Behavioral Intention to Participate in Virtual Communities. *CyberPsychol. Behav.* **2006**, *9*, 5. [CrossRef]
45. Anglin, A.H.; Kincaid, P.A.; Short, J.C.; Allen, D.G. Role theory perspectives: Past, present, and future applications of role theories in management research. *J. Manag.* **2022**, *48*, 1469–1502. [CrossRef]
46. Turel, O.; Connelly, C.E.; Fisk, G.M. Service with an e-smile: Employee authenticity and customer use of web-based support services. *Inf. Manag.* **2013**, *50*, 98–104. [CrossRef]
47. Venkatesh, V.; Davis, F.D. A theoretical extension of the technology acceptance model: Four longitudinal field studies. *Manag. Sci.* **2000**, *46*, 186–204. [CrossRef]
48. Matute, J.; Polo-Redondo, Y.; Utrillas, A. The influence of EWOM characteristics on online repurchase intention: Mediating roles of trust and Perceived Usefulness. *Online Inf. Rev.* **2016**, *40*, 1090–1110. [CrossRef]
49. Filieri, R.; Hofacker, C.F.; Alguezaui, S. What makes information in online consumer reviews diagnostic over time? The role of review relevancy, factuality, currency, source credibility, and ranking score. *Comput. Hum. Behav.* **2018**, *80*, 122–131. [CrossRef]
50. Ryan, R.M.; Deci, E.L. Intrinsic and extrinsic motivation from a self-determination theory perspective: Definitions, theory, practices, and future directions. *Contemp. Educ. Psychol.* **2020**, *61*, 101860. [CrossRef]
51. Jenneboer, L.; Herrando, C.; Constantinides, E. The Impact of Chatbots on Customer Loyalty: A Systematic Literature Review. *J. Theor. Appl. Electron. Commer. Res.* **2022**, *17*, 212–229. [CrossRef]
52. Ruggiero, T.E. Uses and gratifications theory in the 21st century. *Mass Commun. Soc.* **2000**, *3*, 3–37. [CrossRef]
53. Liang, L.J.; Choi, H.S.C.; Joppe, M. Understanding repurchase intention of Airbnb consumers: Perceived Authenticity, electronic word-of-mouth, and price sensitivity. *J. Travel Tour. Mark.* **2018**, *35*, 73–89. [CrossRef]
54. Schnall, R.; Higgins, T.; Brown, W.; Carballo-Dieguez, A.; Bakken, S. Trust, Perceived Risk, Perceived Ease of Use and Perceived Usefulness as Factors Related to mHealth Technology Use. *Stud. Health Technol. Inform.* **2015**, *216*, 467–471. Available online: https://ebooks.iospress.nl/volumearticle/40252 (accessed on 8 August 2024).
55. Hassan, M.; Iqbal, M.S.; Habibah, U. Self-Service Technology Service Quality: Building Loyalty and Intention Through Technology Trust in Pakistani Service Sector. *SAGE Open* **2020**, *10*, 2158244020924412. [CrossRef]
56. Mitchell, V.-W. Consumer Perceived Risk: Conceptualisations and models. *Eur. J. Mark.* **1999**, *33*, 163–195. [CrossRef]
57. Lim, N. Consumers' Perceived Risk: Sources versus consequences. *Electron. Commer. Res. Appl.* **2003**, *2*, 216–228. [CrossRef]
58. Kaur, S.; Arora, S. Role of Perceived Risk in online banking and its impact on behavioral intention: Trust as a moderator. *J. Asia Bus. Stud.* **2020**, *15*, 1–30. [CrossRef]
59. Dowling, G.R. Perceived Risk: The concept and its measurement. *Psychol. Mark.* **1986**, *3*, 193–210. [CrossRef]
60. Smith, V.L. Reconnecting Modern Economics with Its Classical Origins: Discovering Adam Smith. In *A Life of Experimental Economics, Volume II*; Springer: Berlin/Heidelberg, Germany, 2018; pp. 189–215. [CrossRef]
61. Baker, M.J. *Marketing: Critical Perspectives on Business and Management*; Taylor & Francis: Abingdon, UK, 2001.
62. Yao, H.; Zhong, C. The Analysis of Influencing Factors and Promotion Strategy for the Use of Mobile Banking. *Can. Soc. Sci.* **2011**, *7*, 2.
63. Lu, H.; Hsu, C.; Hsu, H. An empirical study of the effect of Perceived Risk upon intention to use online applications. *Inf. Manag. Comput. Secur.* **2005**, *13*, 106–120. [CrossRef]
64. Ara, A.; Thomas, A. Exploring Consumer Perceptions and Ethical Considerations in AI-Powered E-Commerce. In *Data Visualization Tools for Business Applications*; IGI Global: Hershey, PA, USA, 2025; pp. 347–368. [CrossRef]
65. Featherman, M.S.; Pavlou, P.A. Predicting e-services adoption: A Perceived Risk facets perspective. *Int. J. Hum.-Comput. Stud.* **2003**, *59*, 451–474. [CrossRef]
66. Zarouali, B.; Van den Broeck, E.; Walrave, M.; Poels, K. Predicting consumer responses to a chatbot on Facebook. *Cyberpsychol. Behav. Soc. Netw.* **2018**, *21*, 491–497. [CrossRef]
67. Gefen, D.; Karahanna, E.; Straub, D.W. Trust and TAM in online shopping: An integrated model. *MIS Q.* **2003**, *27*, 51–90. [CrossRef]
68. Cova, B.; Shankar, A. Tribal Marketing. In *Marketing Management*; Routledge: London, UK, 2020; pp. 168–182.

69. Canavari, M.; Cantore, N.; Castellini, A.; Pignatti, E.; Spadoni, R. *International Marketing and Trade of Quality Food Products*; Wageningen Academic Publishers: Wageningen The Netherlands, 2009. [CrossRef]
70. Phang, C.W.; Li, Y.; Sutanto, J.; Kankanhalli, A. Senior Citizens' Adoption of E-Government: In Quest of the Antecedents of Perceived Usefulness. In Proceedings of the 38th Annual Hawaii International Conference on System Sciences, Big Island, HI, USA, 3–6 January 2005; p. 130a. [CrossRef]
71. Alexandrakis, D.; Chorianopoulos, K.; Tselios, N. Older Adults and Web 2.0 Storytelling Technologies: Probing the Technology Acceptance Model through an Age-Related Perspective. *Int. J. Hum.–Comput. Interact.* **2020**, *36*, 1623–1635. [CrossRef]
72. Yi, M.Y.; Jackson, J.D.; Park, J.S.; Probst, J.C. Understanding Information Technology Acceptance by Individual Professionals: Toward an Integrative View. *Inf. Manag.* **2006**, *43*, 350–363. [CrossRef]
73. Chu, C.; Lu, H. Factors Influencing Online Music Purchase Intention in Taiwan: An Empirical Study Based on the Value--Intention Framework. *Internet Res.* **2007**, *17*, 139–155. [CrossRef]
74. Zheng, X.; Men, J.; Xiang, L.; Yang, F. Role of Technology Attraction and Parasocial Interaction in Social Shopping Websites. *Int. J. Inf. Manag.* **2020**, *51*, 102043. [CrossRef]
75. Hess, T.J.; McNab, A.L.; Basoglu, K.A. Reliability Generalization of Perceived Ease of Use, Perceived Usefulness, and Behavioral Intentions. *MIS Q.* **2014**, *38*, 1–28. Available online: https://www.jstor.org/stable/26554866 (accessed on 7 September 2024). [CrossRef]
76. Ramos de Luna, I.; Montoro-Ríos, F.; Molinillo, S.; Liébana-Cabanillas, F. Consumer Behaviour and Mobile Payments in the Point of Sale: Exploring the Determinants of Intention to Adopt It. *Int. J. Hum.–Comput. Interact.* **2023**, *40*, 5350–5372. [CrossRef]
77. Bhaskar, P.; Misra, P.; Chopra, G. Shall I Use ChatGPT? A Study on Perceived Trust and Perceived Risk Towards ChatGPT Usage by Teachers at Higher Education Institutions. *Int. J. Inf. Learn. Technol.* **2024**, *41*, 428–447. [CrossRef]
78. Silva, S.C.; De Cicco, R.; Vlačić, B.; Elmashhara, M.G. Using Chatbots in E-Retailing–How to Mitigate Perceived Risk and Enhance the Flow Experience. *Int. J. Retail Distrib. Manag.* **2022**, *51*, 285–305. [CrossRef]
79. Lin, Y.; Marjerison, R.K.; Kennedyd, S. Reposting Inclination of Chinese Millennials on Social Media: Consideration of Gender, Motivation, Content and Form. *J. Int. Bus. Cult. Stud.* **2019**, *12*, 20.
80. Gosling, S.D.; Vazire, S.; Srivastava, S.; John, O.P. Should We Trust Web-Based Studies? A Comparative Analysis of Six Preconceptions About Internet Questionnaires. *Am. Psychol.* **2004**, *59*, 93. Available online: https://psycnet.apa.org/buy/2004-11287-002 (accessed on 7 August 2024). [CrossRef]
81. Rauschnabel, P.A.; Rossmann, A.; tom Dieck, M.C. An Adoption Framework for Mobile Augmented Reality Games: The Case of Pokémon Go. *Comput. Hum. Behav.* **2017**, *76*, 276–286. [CrossRef]
82. Childers, T.L.; Carr, C.L.; Peck, J.; Carson, S. Hedonic and Utilitarian Motivations for Online Retail Shopping Behavior. *J. Retail.* **2001**, *77*, 511–535. [CrossRef]
83. Ko, H.; Cho, C.-H.; Roberts, M.S. Internet Uses and Gratifications: A Structural Equation Model of Interactive Advertising. *J. Advert.* **2005**, *34*, 57–70. [CrossRef]
84. Moon, J.-W.; Kim, Y.-G. Extending the TAM for a World-Wide-Web Context. *Inf. Manag.* **2001**, *38*, 217–230. [CrossRef]
85. Venkatesh, V.; Thong, J.Y.; Xu, X. Unified Theory of Acceptance and Use of Technology: A Synthesis and the Road Ahead. *J. Assoc. Inf. Syst.* **2016**, *17*, 328–376. [CrossRef]
86. Hayes, A.F. *Introduction to Mediation, Moderation, and Conditional Process Analysis: A Regression-Based Approach*; Guilford Publications: New York, NY, USA, 2017.

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